

wave all equations ↓

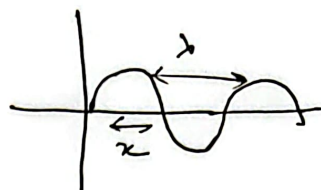
$$y = A \sin \omega t$$

wave speed, $v = f\lambda$

angular freq $\omega, = 2\pi f / \frac{2\pi}{T}$

Time period $T = \frac{1}{f}$

wave length $\lambda = vT$

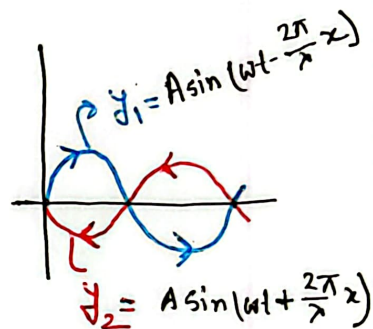


① equation of displacement from ① $y = A \sin(\omega t - \phi)$

$$\Rightarrow y = A \sin \frac{2\pi}{\lambda} (vt - x)$$

② $y_1 = A \sin(\omega t + \phi)$

$$y = A \sin \frac{2\pi}{\lambda} (vt + x)$$



③ differential equation for wave motion

$$y = A \sin \frac{2\pi}{\lambda} (vt - x)$$

$$\frac{dy}{dx} = A \cos \frac{2\pi}{\lambda} (vt - x) \frac{dy}{dx} \left(\frac{2\pi}{\lambda} (vt - x) \right)$$

$$\therefore \frac{dy}{dx} = -\frac{2\pi}{\lambda} A \cos \frac{2\pi}{\lambda} (vt - x)$$

④ Particle velocity, $v_p = -v \cdot dy/dx$

$$= -v \cdot \left(-\frac{2\pi}{\lambda} A \cos \frac{2\pi}{\lambda} (vt - x) \right)$$

⑤ resultant, $y = y_1 + y_2$

$$y = 2A \cos \frac{2\pi}{\lambda} x \sin \omega t$$

①

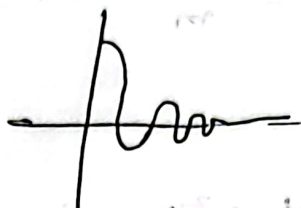
$$\frac{\omega}{\gamma} = 10$$

$$\omega = 10\gamma$$

$$\omega^2 = 100\gamma^2$$

$$\omega^2 = \frac{4 \cdot 100}{4} \times \frac{\gamma^2}{4}$$

$$\therefore \omega^2 > \frac{\gamma^2}{4}$$



$$\frac{\omega}{\gamma} = 0.5$$

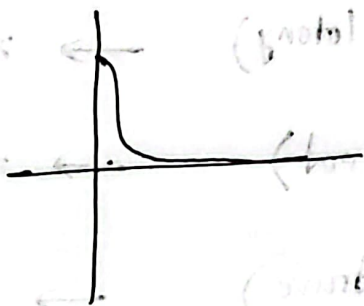
$$\omega = 0.5\gamma$$

$$\omega^2 = 0.25\gamma^2$$

$$\omega^2 = \frac{0.25 \times 4}{4} \gamma^2$$

$$\omega^2 = \frac{1}{4} \gamma^2$$

$$\omega^2 = \frac{\gamma^2}{4}$$



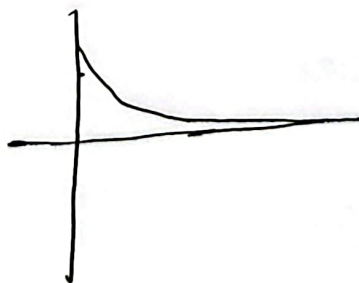
$$\frac{\omega}{\gamma} = 0.03$$

$$\omega^2 = (0.03)^2 \gamma^2$$

$$\omega^2 = \frac{4(0.03)^2}{4} \gamma^2$$

$$\omega^2 = 9 \times 10^{-4} \cdot \frac{\gamma^2}{4}$$

$$\therefore \omega^2 < \frac{\gamma^2}{4}$$



②.

$$L = 0.4 \text{ H}, C = 0.0020 \mu\text{F}.$$

$$= 0.0020 \times 10^{-6} \text{ F}$$

$$\frac{1}{Lc} = \frac{R^2}{4L^2}$$

$$R^2 Lc = 4L^2$$

$$R = \sqrt{\frac{4L^2}{Lc}} = \sqrt{\frac{4 \times (0.4)^2}{0.4 \times 0.0020 \times 10^{-6}}}$$

$$= 28284.27 \Omega$$

$$\textcircled{1} \omega_0 = \frac{1}{\sqrt{Lc}}$$

$$2\pi f = \frac{1}{\sqrt{Lc}}$$

$$f = \frac{1}{2\pi \sqrt{Lc}}$$

$$= \frac{1}{2\pi \sqrt{0.4 \times 0.0020 \times 10^{-6}}}$$

$$\textcircled{3} C = 0.1 \mu\text{F} = 0.1 \times 10^{-6}$$

$$L = 10 \text{ mH} = 10 \times 10^{-3} \text{ H}, R = 200 \Omega$$

$$\frac{1}{Lc} = \frac{1}{10 \times 10^{-3} \times 0.1 \times 10^{-6}}$$

$$= 1 \times 10^9$$

$$\frac{R^2}{4L^2} = \frac{200^2}{4(10 \times 10^{-3})^2}$$

$$= 1 \times 10^8$$

Oscillatory freq $\therefore \frac{1}{Lc} > \frac{R^2}{4L^2}$

$$\therefore \omega_1 = \sqrt{\frac{1}{Lc} - \frac{R^2}{4L^2}}$$

$$= 2\pi f_1 = \sqrt{\frac{1}{Lc} - \frac{R^2}{4L^2}} \Rightarrow f_1 = \frac{\sqrt{\frac{1}{Lc} - \frac{R^2}{4L^2}}}{2\pi}$$

$$= 4774.65 \text{ Hz}$$

Resonant freq

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$2\pi f = \frac{1}{\sqrt{LC}}$$

$$f = \frac{1}{2\sqrt{LC}} = \frac{1}{2\sqrt{10 \times 10^{-6} \times 0.1 \times 10^{-3}}} = 56.32 \text{ Hz}$$

$$\textcircled{4} \textcircled{1} \omega_1 = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

$$\Rightarrow \frac{2\pi}{T} = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}}$$

$$\Rightarrow T = \frac{2\pi}{\sqrt{\frac{k}{m} - \frac{\gamma^2}{4}}}$$

$$= 0.34 \text{ sec}$$

$$k = 85 \text{ N/m}$$

$$m = 250 \text{ gm} = 0.25 \text{ kg}$$

$$b = 70 \text{ gms} = 0.07 \text{ kg/s}$$

$$\gamma = \frac{b}{m} = \frac{0.07}{0.25} = 0.28$$

$$\frac{k}{m} = \frac{85}{0.25} = 340$$

$$\textcircled{11} x = Ae^{-\frac{\gamma t}{2}}$$

$$\frac{A}{2} = Ae^{-\frac{\gamma t}{2}}$$

$$\Rightarrow \frac{1}{2} = e^{-\frac{\gamma t}{2}}$$

$$\Rightarrow -\frac{\gamma t}{2} = \ln\left(\frac{1}{2}\right)$$

$$\Rightarrow t = -\frac{2 \ln\left(\frac{1}{2}\right)}{\gamma}$$

$$= 4.95 \text{ s}$$

$$x = \frac{A}{2} \left[1 \right]$$

~~lifetime~~

$$\text{Life time} = \frac{1}{\gamma} = \frac{1}{0.28} = 3.57$$

$$f = \frac{N}{t}$$

$$N = f t$$

$$\therefore N = \frac{1}{T} t$$

$$= \frac{1}{0.34} \times 3.57$$

$$= 10.5$$

$$t = 20 \times 0.35$$

(iv) Life time = $\frac{1}{\gamma} = 3.57$

(v) $x_{\max} = 0.35 \text{ cm}$

Damping energy $E = \frac{1}{2} k A^2 e^{-\frac{\gamma t}{m}}$

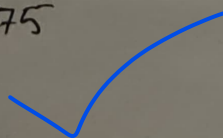
$$= \frac{1}{2} 85 \times (0.35)^2 \times \left(e^{-\frac{0.28 \times 20 \times 0.35}{0.25}} \right)$$

$$= 0.733$$

(vi) $A = A e^{-\frac{\gamma t}{2}}$

$$\frac{A}{A} = e^{-\frac{0.28 \times 20 \times 0.35}{2}}$$

$$= 0.375$$



$$(5) \quad y = 4 \sin(0.1x - 2t) \quad y = A \sin \frac{2\pi}{\lambda} (vt - x) \quad (1)$$

$$= 4 \sin 0.1 \left(\frac{2t}{0.1} - \frac{0.1}{0.1} x \right)$$

$$(i) \text{ Amplitude } A = -4$$

$$(ii) \text{ wavelength } \lambda, \rightarrow \frac{2\pi}{\lambda} = \frac{0.1}{0.1} \Rightarrow \lambda = \frac{2\pi}{0.1} = 20\pi$$

$$(iii) \text{ speed } v = \frac{2}{0.1} \text{ ms}^{-1} = 20 \text{ ms}^{-1}$$

$$(iv) \text{ frequency } f = \frac{v}{\lambda} = \frac{20}{20\pi} = \frac{1}{\pi} \text{ Hz}$$

$$= 0.318 \text{ Hz}$$

$$(6) \quad y = 4.0 \sin 2\pi \left(\frac{x}{100} \right)$$

$$y = 4 \sin \frac{2\pi}{100} \left(\frac{30}{100} \times 3 - x \right)$$

$$v = 30 \text{ cm/s} = 0.3 \text{ m/s}$$

$$(7) \quad y = 5.0 \sin \pi (360t - 0.15x) \rightarrow y = 5 \sin \frac{2\pi}{2} \left(\frac{360}{0.15} t - x \right)$$

$$(i) \text{ Amplitude } = 5$$

$$(ii) \quad v = \frac{360}{0.15} = 2400 \text{ ms}^{-1}$$

$$(iii) \quad \lambda = \frac{2}{0.15} = 13.33$$

$$(iv) \quad T = \frac{1}{f} = \frac{1}{180.04}$$

$$= 5.55 \times 10^{-3}$$

$$(v) \quad f = \frac{2400}{13.33}$$

$$= 180.04 \text{ Hz}$$

$$\textcircled{8} \quad A = 8 \text{ m}$$

$$x_{10} \rightarrow y_6$$

$$x_{25} \rightarrow y_4$$

$$y = A \sin \frac{2\pi}{\lambda} (vt - x)$$

$$y \leftarrow 6 = 8 \sin \frac{2\pi}{\lambda} (vt - 10)$$

$$\sin^{-1}\left(\frac{6}{8}\right) = \frac{2\pi}{\lambda} (vt - 10)$$

$$\frac{2\pi}{\lambda} (vt - 10) = 0.848 \quad \text{--- (i)}$$

$$\frac{2\pi}{\lambda} (vt - 25) = \frac{\pi}{6} \quad \text{--- (ii)}$$

$$\frac{2\pi}{\lambda} vt - \frac{2\pi}{\lambda} 10 = 0.848$$

$$\frac{2\pi}{\lambda} vt - \frac{2\pi}{\lambda} 25 = \frac{\pi}{6}$$

⑧ $A = 8 \text{ m}$ $(x \text{ } 1.0 \text{ } 1.0) \text{ } \pi/2 \text{ } = 0$ (3) $(1.0 \times 1.0) \text{ } \pi/2 \text{ } = 0$

$x_{10} \rightarrow y_6$

$x_{25} \rightarrow y_4$

$(x \text{ } 1.0 \text{ } 1.0) \text{ } \pi/2 \text{ } = 0$

$y = A \sin \frac{2\pi}{\lambda} (vt - x)$

$p = A \sin \frac{2\pi}{\lambda} (vt - x)$ (I)

$6 = 8 \sin \frac{2\pi}{\lambda} (vt - 10)$

$\lambda = 1.0 \text{ m}$ (II)

$\sin^{-1}(\frac{6}{8}) = \frac{2\pi}{\lambda} (vt - 10)$

$v = 1.0 \text{ m/s}$ (III)

$\frac{2\pi}{\lambda} (vt - 10) = 0.848$ — (IV)

$\frac{2\pi}{\lambda} (vt - 25) = \frac{\pi}{6}$ — (V)

$(\frac{x}{100}) \text{ } \pi/2 \text{ } = 0$ (VI)

$\frac{2\pi}{\lambda} vt - \frac{2\pi}{\lambda} 10 = 0.848$

$\frac{x}{100} \text{ } \pi/2 \text{ } = 0$

$\frac{2\pi}{\lambda} vt - \frac{2\pi}{\lambda} 25 = \frac{\pi}{6}$

$(x \text{ } 1.0 \text{ } 1.0) \text{ } \pi/2 \text{ } = 0$ (VII)

$\frac{20\pi}{\lambda} + \frac{50\pi}{\lambda} = 0.848 - \frac{\pi}{6}$

$p = A \sin \frac{2\pi}{\lambda} (vt - x)$ (I)

$\frac{-20\pi + 50\pi}{\lambda} = 0.324$

$v = 1.0 \text{ m/s}$ (II)

$\frac{30\pi}{\lambda} = 0.324$

$\lambda = 1.0 \text{ m}$ (III)

$\lambda = \frac{30\pi}{0.324} = 290$

$\frac{50\pi}{\lambda} = 0.324$ (IV)

⑨ $T = 5.00 \text{ s.}$

$\omega = 2\pi f$

$\omega^2 = \frac{g}{L}$

$\frac{2\pi}{T} = \sqrt{\frac{g}{L}}$

$T = 2\pi \sqrt{\frac{L}{g}}$

$T^2 = 4\pi^2 \cdot \frac{L}{g}$

$L 4\pi^2 = g T^2$

$L = \frac{g T^2}{4\pi^2} = 6.20 \text{ m}$

$f = \frac{1}{T} = 0.2 \text{ Hz}$

⑩ resultant $y = y_1 + y_2$

$= A \cos\left(\frac{1}{3} kx + \omega t\right) + A \cos\left(\frac{1}{3} kx - \omega t\right)$

$= A (\cos\left(\frac{1}{3} kx + \omega t\right) + \cos\left(\frac{1}{3} kx - \omega t\right))$

$= 2A \cos \frac{\frac{1}{3} kx + \omega t + \frac{1}{3} kx - \omega t}{2} \cos \frac{\frac{1}{3} kx + \omega t - \frac{1}{3} kx + \omega t}{2}$

$= 2A \cos \frac{2 \cdot \frac{1}{3} kx}{2} \cos \frac{2\omega t}{2}$

$= 2A \cos \frac{1}{3} kx \cos \omega t$

node

$$\cos\left(\frac{1}{3}kx\right) = 0$$

$$[0 \pi \dots \pi]$$

$$\frac{1}{3}kx = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$$

$$\frac{kx}{2} = \frac{\pi}{2}$$

$$3\pi = 2kx$$

$$x = \frac{3\pi}{2k}$$

$$k = \frac{2\pi}{\lambda}$$

$$x = \frac{3\pi}{2 \times \frac{2\pi}{\lambda}} = \frac{3\pi}{\frac{4\pi}{\lambda}} = \frac{3\lambda}{4}$$

$$\cos\left(\frac{1}{3}kx\right) = 1$$

antinode =

$$\cos\left(\frac{1}{3}kx\right) = 1$$

$$[1 \text{ to } \pi \text{ Max}]$$

$$\frac{1}{3}kx = \cos^{-1}(1)$$

$$\frac{1}{3}kx = 0, \pi, 2\pi, 3\pi \dots$$

11

$$(11) \quad x = A e^{-\alpha t} \cos(\omega_d t + \phi)$$

$$v = -A \omega_d e^{-\alpha t} \sin(\omega_d t + \phi)$$

$$E = \frac{1}{2} m v^2 + \frac{1}{2} k x^2$$

$$= \frac{1}{2} m \omega_d^2 A^2 e^{-2\alpha t} \sin^2(\omega_d t + \phi) + \frac{1}{2} k A^2 e^{-2\alpha t} \cos^2(\omega_d t + \phi)$$

$$= \frac{1}{2} k A^2 e^{-2\alpha t}$$

$$= \frac{1}{2} k A^2 e^{-2\alpha t}$$

$$= E_0 e^{-2\alpha t}$$

shunt

$$\begin{aligned} \omega^2 &= \frac{k}{m} \\ m \omega^2 &= k \\ m &= \frac{k}{\omega^2} \end{aligned}$$

12) $v_1 = c_1 = 2 \text{ cm/s} = 0.02 \text{ ms}^{-1}$

$$\lambda_1 = 4 \text{ cm} = 0.04 \text{ ms}$$

$$v_2 = c_2 = 3 \text{ cm}$$

$$\lambda_2 = 3 \text{ cm}$$

$$\begin{aligned} \omega &= 2\pi f \\ &= 2\pi \frac{v}{\lambda} \\ v &= f \lambda \end{aligned}$$

$$\omega_1 = 2\pi \frac{v_1}{\lambda_1}$$

$$\omega_2 = 2\pi \frac{v_2}{\lambda_2}$$

$$= 2\pi \frac{2}{4}$$

$$= 2\pi \frac{3}{3}$$

$$= \pi$$

$$= 2\pi$$

